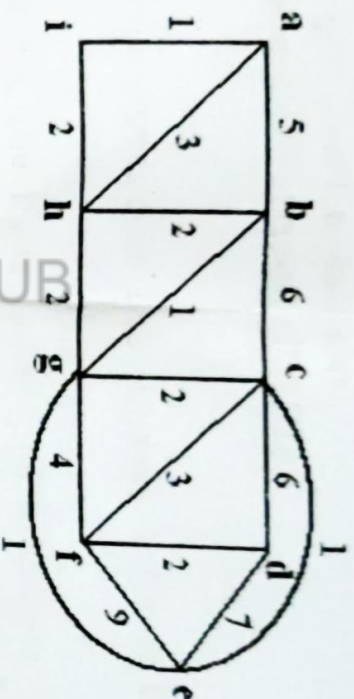


- (b) Use Kruskal's algorithm to find a minimum spanning tree in the following weighted graph: (06)



- (c) Use Huffman coding to encode these symbols with given frequencies:
A: 0.15, B: 0.20, C: 0.05, D: 0.10, E: 0.30, F: 0.08, G: 0.09, H: 0.03
 What is the average number of bits required to encode a symbol? (06)

- Q5. (a) Show that the set of all integers $\mathbb{I}$ forms an infinite abelian group with the operation $\cdot$ defined as: $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} + \mathbf{b} - 1$, where all $\mathbf{a}, \mathbf{b} \in \mathbb{I}$. (06)

- (b) Show that the set of numbers of the form $\mathbf{a} + \mathbf{b}\sqrt{\mathbf{m}}$, where $\mathbf{m}$ is a positive integer and with $\mathbf{a}$ and $\mathbf{b}$ as rational numbers, is a field. (06)

- (c) (i) Transform $\mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = ((\overline{\mathbf{x} \cdot \mathbf{y}}) \cdot \overline{(\overline{\mathbf{x} + \mathbf{z}}) \cdot (\overline{\mathbf{y} + \mathbf{z}})})$ into a sums-of-products form. (03)
- (ii) Simplify the Boolean function and then draw the simplified logic circuit and switching circuit: (03)

$$\mathbf{f}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} + \mathbf{c}) \cdot (\overline{\mathbf{c} + \mathbf{a}})$$